

Comparison of Naive Bayes and Decision Tree for Aspect-Based Sentiment Analysis on Indonesian Restaurant Reviews

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Abstract—Product reviews that can be accessed online help shoppers get information about the products they are about to buy. The emergence of e-commerce websites has made it possible for customers to express their opinions on goods and services. Business owners can gain an understanding of customer satisfaction. The problem that arises is that a large number of reviews can overwhelm users to read all the reviews, requiring sentiment analysis. Sentiment analysis is done to overcome this problem, by summarizing reviews into sentiments and displaying the sentiments of these reviews. To perform sentiment analysis on restaurant reviews, a classification method is needed. Naive Bayes is one of the classification methods that can be used to classify these sentiments. Decision Tree is used to compare the classification results of the Naive Bayes method. Both methods were chosen because they can produce a good classification model with a relatively simple process and relatively little data. The process of both classification methods uses a learning process and a testing process to obtain classification results. The experimentation resulted in Gaussian Naive Bayes that gave the best performance with the best f1-score of 90.81%. Compared with the Gini Decision Tree with the best f1-score of 89.64%.

Index Terms—naive bayes, decision tree, sentiment analysis, NLP

I. INTRODUCTION

Individuals typically consult the consensus before making a choice. Before making a purchase, they frequently seek the advice of friends and relatives. The emergence of e-commerce websites in the last ten years has made it possible for customers to express their opinions on goods and services. Business owners might give their clients a space to post evaluations of the goods or services they have received. Business owners may learn more about what makes customers happy. Before purchasing a product, prospective customers can use reviews to help them make a better choice. However, consumers may find it overwhelming to read and consider every review [1]. As a result, reviewing reviews requires sentiment analysis.

Sentiment analysis refers to the classification of text based on the sentiment it contains. Sentiment analysis is also known as opinion mining, opinion extraction, and influence analysis in the literature. Sentiment analysis has received considerable attention from companies' leading IT experts because by understanding public opinion, they can improve customer satisfaction [2]. However, problems can occur because reviews can contain biased information, diverse information, and incomplete information.

To perform sentiment analysis on restaurant reviews, a classification method is needed to predict the sentiment

of said reviews. Extracting review information can lead to several issues. These issues are ones of uncertainty, and classifiers and other machine learning techniques can help reduce them. Numerous efficient algorithms for classification have been created, such as Decision Tree and Naive Bayes [3]. Naive Bayes is an efficient classification method. It requires only one scan of the training data. Naive Bayes can also compete well with more complicated classifiers [4]. However, given the independence assumption, it can limit its applicability in some cases [5]. Decision Tree is one of the popular classification algorithms. It is a prediction model with a tree structure where each internal node represents a test on an attribute. Each branch on the Decision Tree represents a test result and a class or class distribution labels each leaf node [4]. Decision trees are simple and easy to understand. However, it is prone to overfitting, particularly when the tree is not limited in height or not pruned [5].

It has been demonstrated that Naive Bayes and Decision trees perform admirably in a variety of classification tasks, including sentiment analysis. One such is the study conducted by Guia et al. [6], which compares the effectiveness of Random Forest, Naive Bayes, Support Vector Machines, and Decision Trees in the classification of product reviews. The outcomes demonstrated that Naive Bayes and Decision Trees are competitive with the more complex methods. As a result, the performance of Decision Trees and Naive Bayes will be compared in this study.

There are three processes involved in this research. They include preprocessing, feature extraction, and classification of aspects and their sentiments. Preprocessing aimed at cleaning and preparing the data for the second step is feature extraction. CountVectorizer, known as term frequency, calculates how often a word appears in a document. It keeps track of the frequency of each unique token in the document; the higher the token's score, the more often it occurs in the text [7]. Then, the results are used as a model to create Decision Tree and Naive Bayes models. Then, Naive Bayes and Decision Tree will classify the aspects and sentiment. Finally, in this research, both will be compared to get the optimal method for sentiment analysis.

II. RELATED WORKS

This research uses references from previous studies that are related to aspect-based sentiment analysis, sentiment analysis, classification methods, and evaluation. Sentiment

analysis includes various sub-tasks, and many studies have been devoted to this research area.

Mubarak et al. [8] discussed aspect-based sentiment analysis on the restaurant review dataset from SemEval 2014 Task 4. This study used Naïve Bayes with Chi-Square and Part-of-Speech Tagging (POS Tagging) as additional features in the classification process. Naïve Bayes was chosen because the method is trusted to solve uncertainty problems that arise in sentiment analysis tasks. The classification performance obtained a value in the form of an f1-score of 78.12%.

Guia et al. [6] researched sentiment analysis of Amazon mobile phone reviews. This research compares Naive Bayes, Support Vector Machine, Decision Tree, and Random Forest. It also studied the impact of brand and price attributes on the polarity of the reviews. This research resulted in Decision Tree classification obtaining the best f1-score of 82% and Bernoulli Naive Bayes obtaining the best f1-score of 84%.

Widhianto and Romadhony [9] researched aspect-based sentiment analysis on the same dataset. This research uses deep learning methods with CNN and LSTM classification methods. This research also implements N-grams and CBOW as additional features and IOB2 format. This research obtained the best micro average f1-score of 55.1% with LSTM and 52.2% with CNN. In contrast to this research, this study will explore the effectiveness of a less complex classification and sampling method to tackle the imbalance problem found in Widhianto and Romadhony's research.

Singgalen [10] discussed sentiment analysis of restaurant review datasets using several classification methods. The classification methods included are Decision Tree and Naive Bayes. The classification performance obtained an accuracy of 93.23% for Decision Tree and 93.23% for Naive Bayes. This study found that consumers have more positive sentiments towards cuisine, product, service, price, restaurant condition, and business location.

Compared to the four mentioned studies, this research uses Naive Bayes and Decision Trees paired with CountVectorizer and RandomOversampling along with comparing Naive Bayes as a probability-based classification to Decision Trees as an entropy-based classification.

III. METHODOLOGY

The system to be built in this research is a system that analyzes the sentiment of restaurant reviews. Fig. 1 shows the design of the system to be built.

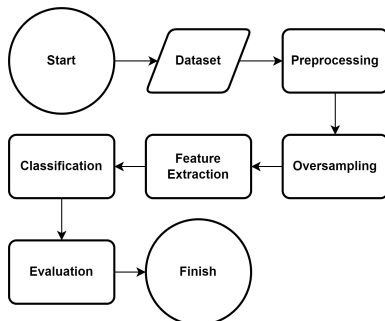


Fig. 1. System design flowchart.

A. Dataset

The Zomato website's restaurant reviews for Bandung, Indonesia, which were retrieved via Dataverse [11], served as

the dataset for this study. There are 4798 Indonesian-language review data in the collection.

Out of 4798 reviews in the dataset, 1495 records do not have any aspect, therefore, said records are deleted to remove any inconsistencies in this research. The dataset used in this research consists of user reviews, aspects of said reviews, and their sentiments. Out of 3303 records, individuals tend to review more about the FOOD aspect of a restaurant.

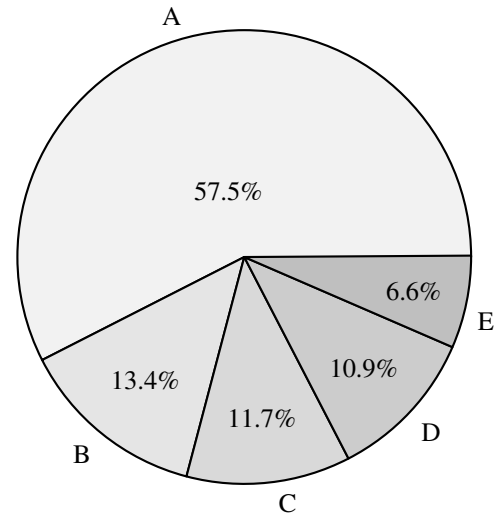


Fig. 2. Aspect Distribution.

Fig. 2 shows the distribution of all aspects in the dataset. Where *A* is FOOD with 1900 records, *B* is MISCELLANEOUS with 441 records, *C* is AMBIENCE with 385 records, *D* is SERVICE with 359 records, and *E* is PRICE with 218 records. Data are then divided with a ratio of 7:3 for train and test respectively.

B. Preprocessing

Before the data was ready to train the classification algorithm, it underwent the preprocessing step. Fig. 3 shows the steps that the data went through.

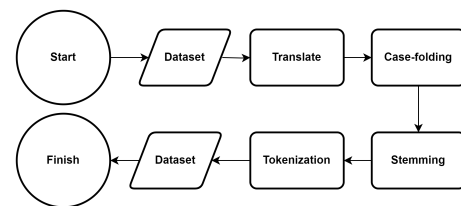


Fig. 3. Preprocessing flowchart.

1) *Translate*: The data gets translated from other languages, such as English to Indonesian. Translation is done to convey the original meaning of a text in a different language. An example of a translation can be seen in Table I.

TABLE I
TRANSLATE

Sentence	Translated Sentence
Mushroom sauce taste just like cream sauce, no mushroom.	Saus jamur rasanya seperti saus krim, tanpa jamur.

2) *Case folding*: Case folding is done to remove symbols, punctuation, emojis, and capitalized words. However, case folding may lead to missing information on proper nouns [12]. An example of case folding is as shown in Table II.

TABLE II
CASE FOLDING

Sentence	Case folded Sentence
Saus jamur rasanya seperti saus krim, tanpa jamur.	saus jamur rasanya seperti saus krim tanpa jamur

3) *Stemming*: Stemming reduces inflected forms and changes the word into a common base form. In this research, stemming is done using Sastrawi. It utilizes a Nazief and Adriani-based algorithm, which is improved with the CS (Confix Stripping) algorithm, the ECS Algorithm (Enhanced Confix Stripping), and the Modified ECS algorithm [13]. An example of stemming can be seen in Table III.

TABLE III
STEMMING

Sentence	Stemmed Sentence
saus jamur rasanya seperti saus krim tanpa jamur	saus jamur rasa seperti saus krim tanpa jamur

4) *Tokenization*: Tokenization is done to identify tokens, or basic units that do not need to be deciphered in the next process [14]. Tokenization is done by separating text with space constraints to form tokens. An example of Tokenization can be seen in Table IV.

TABLE IV
TOKENIZATION

Sentence	Stemmed Sentence
saus jamur rasa seperti saus krim tanpa jamur	"saus","jamur","rasa","seperti","saus","krim","tanpa","jamur"

C. Feature Extraction

The data must be transformed into appropriate data types for the selected method to be applied to it and processed. Therefore, feature extraction will be needed.

A fundamental technique for turning text data into a numerical number is CountVectorizer. Following the formation of vectors, a lexicon of the same size and dimension is then obtained. If the data shapes correspond to the same vocabulary word, it will be included in that dimension [15].

D. Oversampling: Random Oversampling

Oversampling is used in this research to avoid an imbalance in the sentiment column of the dataset in this research. It is done using imbalanced-learn's RandomOverSampler (ROS). One of the main advantages of ROS is that it is a simple and easy-to-implement method that does not require any additional information about the data. ROS proves that it is a simple yet competitive sampling method compared to a more complex sampling method such as SMOTE [16]. In contrast to undersampling, oversampling can be used to achieve data balancing by generating new samples every x_{min} to reach the $r_{x_{res}}$ balancing ratio. However, oversampling can lead to overfitting [17].

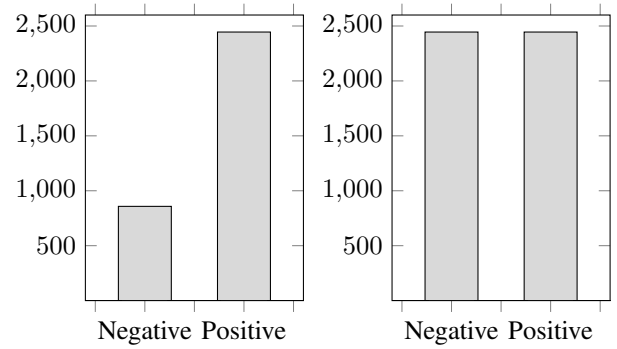


Fig. 4. Sentiment distribution before (left) and after oversampling (right).

This dataset consists of 858 negative sentiments and 2445 positive sentiments, totaling 3303 records. As shown in Fig. 4, it is biased toward positive reviews. This problem can be solved with oversampling. After oversampling, there are 2445 negative sentiments and 2445 positive sentiments.

E. Classification: Naive Bayes

Text classification using the Naive Bayes learning method is based on a probabilistic generative model. The model assumes that each document is generated by a distribution controlled by hidden parameters. The training data is utilized to estimate these hidden parameters, which are then used to classify the documents using Bayes' rule, by computing posterior probabilities. Finally, the classification process involves selecting the most probable class based on the computed probabilities.

Naive Bayes is linear in the number of training examples, which is one of the great strengths of Naive Bayes, i.e., it is very efficient. Equation (1) displays the Naive Bayes formula.

$$P(c|x) = \frac{P(x|c)P(c)}{P(x)} \quad (1)$$

where c is data with a specific class, x is data with an unknown class, $P(c|x)$ is posterior probability of the class given the predictor, $P(x|c)$ is predictor probability of the given class, $P(c)$ is prior probability of the class, and $P(x)$ is prior probability of the predictor [18].

F. Classification: Decision Tree

Decision Tree is a type of sequential model which combines a series of simple tests. These tests evaluate a numeric attribute against a threshold value or a nominal attribute against a set of probable values. This type of symbolic classifier has an advantage over "black-box" models like neural networks, due to its completeness. The logical rules that Decision Trees follow are much easier to interpret compared to the numerical weights of the connections between nodes in a neural network [19]. An attribute is tested by each internal node, the test results are represented by each branch, and the class label is stated by each leaf node in the Decision Tree, a flowchart structure that resembles a tree. Conversely, the root node is the node at the top of the decision tree [20]. There are two splitting criterion used in this research, namely, Entropy and Gini. Equation (2) shows the formula for entropy.

$$Entropy(S) = \sum_{c \in C} -p(c) \log_2 p(c) \quad (2)$$

where S represents the dataset that entropy is calculated, c represents the classes in set S , and $p(c)$ represents the

proportion of data points that belong to class c to the number of total data points in set S . Next, the definition of the Gini Impurity is as shown in (3).

$$Gini(S) = 1 - \sum_{i=1}^k p_i^2 \quad (3)$$

where S represents a dataset that includes examples from k different classes. At a given node, the probability of samples belonging to class i is represented by the symbol p_i [21].

G. Confusion Matrix

Confusion matrix is a table that is created for a classifier operating on a binary dataset, and it is utilized to evaluate the effectiveness of the methods employed [22]. The performance is then evaluated by the data matrix as shown in Table V.

TABLE V
CONFUSION MATRIX

Actual Values	Predicted Values	
	Positive	Negative
Positive	TP	FP
Negative	FN	TN

To evaluate the classification process, it is necessary to calculate the f1-score. f1-score is the average comparison of precision and recall. The f1-score formula is shown in (4).

$$f1\text{-score} = \frac{TP}{TP + \frac{1}{2}(FP + FN)} \quad (4)$$

where TP is True Positive, FP is False Positive, TN is True Negative, and FN is False Negative.

IV. RESULTS AND EXPERIMENTS

In this section, there will be five scenarios that this research will look after. Scenario 1 assessed the potential impact of stemming on classification performance using the baseline dataset, exploring whether improvements would manifest in Naive Bayes and Decision Tree algorithms. Scenario 2, the investigation focused on discerning the effects of oversampling on the best-performing configuration identified in Scenario 1, aiming to determine if this technique could yield a notable increase in accuracy. Scenario 3 compares various Naive Bayes (NB) methods (Gaussian (GNB), Multinomial (MNB), Bernoulli (BNB), and Complement (CNB)) using the best result from Scenario 2 to identify the most effective. Concurrently, Scenario 4 delved into Decision Tree criteria (Gini (GDT) and Entropy (EDT)) to establish the optimal selection for the dataset. Lastly, Scenario 5 sought to compare the outcomes of Scenarios 3 and 4, evaluating whether Naive Bayes or Decision Tree would emerge as more accurate. These scenarios collectively constitute a comprehensive exploration of the potential influences of stemming, oversampling, and algorithmic choices in analyzing restaurant reviews.

A. Scenario 1: Stemming

Using the Naive Bayes and Decision Tree algorithms, this research investigated whether stemming improves text classification performance in the first scenario. Finding out if Stemming has a noticeable impact on classification accuracy for both algorithms.

TABLE VI
VALUE OF ACCURACY AND F1-SCORE OF BOTH METHODS WITH AND WITHOUT STEMMING

Method	Accuracy	F1-score
NB w/ stemming	75.88%	71.66%
DT w/ stemming	81.05%	75.49%
NB	77.83%	74.05%
DT	80.57%	74.64%

As shown in Table VI, both methods are affected differently with stemming. NB performs noticeably worse but DT slightly performs better. This difference in impact is caused by the different underlying assumptions of these algorithms. While stemming can disrupt the independence assumption of NB, it can help decision trees by simplifying the feature space [23].

B. Scenario 2: Oversampling

The second experiment investigated the potential performance improvements through oversampling, specifically from the best-performing configuration identified in Scenario 1. This examination aimed to reveal whether oversampling could further augment the performance of the sentiment analysis models.

TABLE VII
VALUE OF ACCURACY AND F1-SCORE OF DT WITH STEMMING WITH AND WITHOUT OVERSAMPLING

Method	Accuracy	F1-score
DT w/ oversampling	89.64%	89.64%
DT	81.05%	75.49%

Table VII shows that oversampling noticeably increases DT performances by 8%. In the case of DT, oversampling helps by providing more examples of the minority class, which improves the performance of the tree's splits and leads to a better classification performance. Khrisnand and Sangar [16] found that random oversampling significantly increases the performance of decision trees in sentiment analysis.

C. Scenario 3: Naive Bayes Methods

The third scenario focused on comparing various Naive Bayes methods—GNB, MNB, BNB, and CNB—in the context of the best result obtained in Scenario 2 through oversampling.

TABLE VIII
COMPARING NB METHODS WITH STEMMING AND OVERSAMPLING

Method	Accuracy	F1-score
GNB	90.83%	90.81%
MNB	88.65%	90.65%
BNB	86.02%	85.98%
CNB	88.85%	88.85%

GNB performs slightly better than other NB methods with an up to 4.8% lead, as shown in Table VIII. GNB is a variant of NB that assumes that the features are normally distributed, which can be a good assumption for some datasets [5].

D. Scenario 4: Decision Tree Criterion

In Scenario 4, the study turned attention to the comparison of Decision Tree criteria, Gini and Entropy, leveraging the best result obtained in Scenario 2.

TABLE IX

COMPARING DT CRITERION WITH STEMMING AND OVERSAMPLING

Criterion	Accuracy	F1-score
GDT	89.64%	89.64%
EDT	89.51%	89.51%

Table IX shows that GDT performs slightly better than EDT in sentiment analysis, possibly due to its ability to measure impurity and calculate the probability of misclassification [21]. However, the difference in performance between the two criteria is not significant.

E. Scenario 5: Comparing Naive Bayes and Decision Tree

The fifth and final scenario involved a comparative analysis between Naive Bayes and Decision Tree models, incorporating insights from Scenarios 3 and 4.

TABLE X

COMPARING DT AND NB WITH STEMMING AND OVERSAMPLING

Criterion	Accuracy	F1-score
GDT	89.64%	89.64%
EDT	89.51%	89.51%
GNB	90.83%	90.81%
MNB	88.65%	90.65%
BNB	86.02%	85.98%
CNB	88.85%	88.85%

As shown in Table X, GNB performs better with a 1.19% difference. NB is strongly associated with text-based classification, and its independence assumption is a crucial factor for the classifier's success. NB is capable of handling continuous and discrete datasets and can be applied to binary as well as multi-class classification problems. NB works better than the nonparametric Decision Trees model, even though it depends on the independence assumption. In Decision Trees, independent variables do not follow any particular probability distribution [23].

V. CONCLUSION

Compared with Decision Tree, Naive Bayes is better at classifying sentiment within restaurant reviews, with a slight but still comparable lead with GNB with stemming with 90.83% accuracy and 90.81% f1-score. Even with an imbalance of data, both method shows great results in classifying sentiments in restaurant reviews.

Balancing the data with oversampling greatly boosts all methods' accuracy and f1-score, meaning all methods can get better results with better split-ratio and hyper-parameter tuning. However, even with a high accuracy and f1-score, this study has potential limitations. Both methods were applied to a small dataset and results might differ when applied to a large dataset. Larger datasets might require a more complex method to better classify sentiments within. Moreover, the stemming method used in this study is sufficient for this study, although it is slow and takes a lot of processing time.

This research recommends using cross-validation to explore various data splitting and reduces bias towards one class. Other oversampling or undersampling (or a combination of both) methods can also be used to handle imbalance within the dataset. Since this research's stemming had a different impact, future research can also explore other or newer stemming methods.

REFERENCES

- [1] E. Marrese-Taylor, J. D. Velásquez, and F. Bravo-Marquez, "A novel deterministic approach for aspect-based opinion mining in tourism products reviews," *Expert Systems with Applications*, vol. 41, no. 17, pp. 7764–7775, Dec. 2014. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0957417414003315>
- [2] K. Amarouche, H. Benbrahim, and I. Kassou, "Product Opinion Mining for Competitive Intelligence," *Procedia Computer Science*, vol. 73, pp. 358–365, 2015. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S1877050915034651>
- [3] C. X. Ling and V. S. Sheng, "Cost-Sensitive Learning and the Class Imbalance Problem," *Encyclopedia of Machine Learning*, Jan. 2010.
- [4] J. Wang, Ed., *Encyclopedia of Data Warehousing and Mining, Second Edition*. IGI Global, 2009. [Online]. Available: <http://services.igi-global.com/resolvedoi/resolve.aspx?doi=10.4018/978-1-60566-010-3>
- [5] M. Grancharova and M. Jangefalk, "Comparative study of the combined performance of learning algorithms and preprocessing techniques for text classification," 2018.
- [6] M. Guia, R. R. Silva, and J. Bernardino, "Comparison of naïve bayes, support vector machine, decision trees and random forest on sentiment analysis," *KDIR*, vol. 1, pp. 525–531, 2019.
- [7] M. R. H. Khan, U. S. Afroz, A. K. M. Masum, S. Abujar, and S. A. Hossain, "Sentiment analysis from bengali depression dataset using machine learning," in *2020 11th international conference on computing, communication and networking technologies (ICCCNT)*. IEEE, 2020, pp. 1–5.
- [8] M. S. Mubarak, A. Adiwijaya, and M. D. Aldhi, "Aspect-based sentiment analysis to review products using naïve bayes," in *AIP conference proceedings*, vol. 1867, no. 1. AIP Publishing, 2017.
- [9] R. A. Widhianto and A. Romadhony, "Aspect Term Extraction Using Deep Learning-Based Approach on Indonesian Restaurant Reviews," in *2021 9th International Conference on Information and Communication Technology (ICICT)*. Yogyakarta, Indonesia: IEEE, Aug. 2021, pp. 202–206. [Online]. Available: <https://ieeexplore.ieee.org/document/9527472/>
- [10] Y. A. Singgalen, "Sentiment Analysis on Customer Perception towards Products and Services of Restaurant in Labuan Bajo," *Journal of Information Systems and Informatics*, vol. 4, no. 3, pp. 511–523, Sep. 2022. [Online]. Available: <https://journal-isi.org/index.php/isi/article/view/276>
- [11] A. Romadhony and R. A. Widhianto, "Aspect Based Sentiment Analysis - Indonesian Restaurant Review," Mar. 2022. [Online]. Available: <https://doi.org/10.34820/FK2/UVHFOF>
- [12] Department of Computer Science and Engineering, Maulana Azad National Institute of Technology, Bhopal, 462003, India, S. Bauskar, V. Badole, P. Jain, and M. Chawla, "Natural Language Processing based Hybrid Model for Detecting Fake News Using Content-Based Features and Social Features," *International Journal of Information Engineering and Electronic Business*, vol. 11, no. 4, pp. 1–10, Jul. 2019. [Online]. Available: <http://www.mecs-press.org/ijieeb/ijieeb-v11-n4/v11n4-1.html>
- [13] M. A. Rosid, A. S. Fitriani, I. R. I. Astutik, N. I. Mulloh, and H. A. Gozali, "Improving text preprocessing for student complaint document classification using sastrawi," in *IOP Conference Series: Materials Science and Engineering*, vol. 874, no. 1. IOP Publishing, 2020, p. 012017.
- [14] J. J. Webster and C. Kit, "Tokenization as the Initial Phase in NLP," in *COLING 1992 Volume 4: The 14th International Conference on Computational Linguistics*, 1992.
- [15] S. Vijayaraghavan and D. Basu, "Sentiment analysis in drug reviews using supervised machine learning algorithms," *arXiv preprint arXiv:2003.11643*, 2020.
- [16] U. Krishnan and P. Sangar, "A rebalancing framework for classification of imbalanced medical appointment no-show data," *Journal of Data and Information Science*, vol. 6, no. 1, pp. 178–192, 2021.
- [17] G. Lemaître, F. Nogueira, and C. K. Aridas, "Imbalanced-learn: A python toolbox to tackle the curse of imbalanced datasets in machine learning," *The Journal of Machine Learning Research*, vol. 18, no. 1, pp. 559–563, 2017.
- [18] Q. Ren, H. Zhang, D. Zhang, X. Zhao, L. Yan, J. Rui, F. Zeng, and X. Zhu, "A framework of active learning and semi-supervised learning for lithology identification based on improved naïve bayes," *Expert Systems with Applications*, vol. 202, p. 117278, 2022.
- [19] S. B. Kotsiantis, "Decision trees: a recent overview," *Artificial Intelligence Review*, vol. 39, no. 4, pp. 261–283, Apr. 2013. [Online]. Available: <http://link.springer.com/10.1007/s10462-011-9272-4>
- [20] A. Bayhaqy, S. Sfenrianto, K. Nainggolan, and E. R. Kaburuan, "Sentiment analysis about e-commerce from tweets using decision tree, k-nearest neighbor, and naïve bayes," in *2018 international conference on orange technologies (ICOT)*. IEEE, 2018, pp. 1–6.

- [21] S. Tangirala, "Evaluating the impact of gini index and information gain on classification using decision tree classifier algorithm," *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 2, pp. 612–619, 2020.
- [22] J. Maria Navin and R. Pankaja, "Performance analysis of text classification algorithms using confusion matrix," *International Journal of Engineering and Technical Research (IJETR)*, vol. 6, no. 4, pp. 75–8, 2016.
- [23] K. Yadav and R. Thareja, "Comparing the performance of naive bayes and decision tree classification using r," *International Journal of Intelligent Systems and Applications*, vol. 11, no. 12, p. 11, 2019.